AWS S3 LIFECYCLE ASSIGNMENT

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Subject: Amazon Web Services (AWS)

Topic: S3 Lifecycle Management and Cost Optimization

Q.1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

Answer: I will sign in to the AWS Management Console and open Amazon S3. I will select Create bucket and enter the bucket name **my-zomato-images**. I will choose the required AWS Region and keep Block Public Access enabled. After creating the bucket, I will open it, select Upload, and add at least three sample menu images, such as **pizza.jpg**, **burger.jpg**, and **paneer-tikka.jpg**. Finally, I will click Upload and confirm that all three images appear in the Objects list.

Q.2. Create a lifecycle policy to transition all objects older than 30 days to the Glacier storage class.

Answer: I will open the bucket and go to the Management tab, then select Create lifecycle rule. I will name the rule **Move-all-objects-to-Glacier** and apply it to all objects in the bucket. Under Lifecycle rule actions, I will select Transition current versions of objects between storage classes, choose **Glacier Flexible Retrieval** (S3 Glacier), and set the number of days after object creation to **30**. I will review and save the rule.

Q.3. Estimate and compare the monthly storage cost of keeping 5 GB of images in S3 Standard and S3 Glacier using AWS Pricing Calculator.

Answer: For an example estimate, I will select Amazon S3 in AWS Pricing Calculator, choose the Mumbai Region, enter 5 GB of stored data, and compare S3 Standard with S3 Glacier Flexible Retrieval. The following illustration uses approximate example rates of USD 0.025 per GB-month for S3 Standard and USD 0.0036 per GB-month for Glacier Flexible Retrieval. The final calculator amount can differ based on the selected region, current AWS rates, requests, retrieval, and other usage.

Storage class	Example rate / GB-month	5 GB monthly estimate
S3 Standard	US$0.025	US$0.125 ≈ ■11.88
S3 Glacier Flexible Retrieval	US$0.0036	US$0.018 ≈ ■1.71

Estimated difference: Glacier Flexible Retrieval costs about **■10.17 less per month** for 5 GB of stored data, using an illustrative exchange rate of ■95.04 per US dollar. This is a storage-only estimate, not a guaranteed AWS bill or a verified saved Pricing Calculator quote. Glacier retrieval charges and minimum storage duration can affect the total cost.

Q.4. Add a lifecycle rule to delete objects with the prefix 'offers/' after 60 days.

Answer: I will open the bucket's Management tab and create another lifecycle rule named **Delete-old-offers**. In the rule scope, I will select Limit the scope using one or more filters and enter the prefix **offers/**. Under Lifecycle rule actions, I will select Expire current versions of objects and set the number of days after object creation to **60**. I will review and save the rule. This rule will apply to objects whose keys start with offers/, such as offers/summer2024.jpg.

Q.5. Generate a sample JSON configuration that transitions files to Glacier after 15 days and deletes them after 180 days. Explain each section.

Answer: The following is a sample S3 lifecycle configuration. It applies to all objects in the bucket, transitions eligible objects to the S3 Glacier Flexible Retrieval storage class after 15 days, and expires the current object version after 180 days.

```
{
  "Rules": [
    {
      "ID": "Transition-and-Delete-Objects",
      "Status": "Enabled",
      "Filter": {},
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 180
      }
    }
  ]
}
```

Explanation:

• **Rules:** Contains the lifecycle rules for the bucket.
• **ID:** A name used to identify this rule.
• **Status:** Enabled means AWS applies the rule.
• **Filter:** An empty filter applies the rule to all objects in the bucket.
• **Transitions:** Defines when an object moves to another storage class.
• **Days: 15:** The transition happens 15 days after the object is created.
• **StorageClass: GLACIER:** Moves the object to S3 Glacier Flexible Retrieval.
• **Expiration: Days: 180:** Expires the current object version after 180 days. If bucket versioning is enabled, previous versions need separate noncurrent-version expiration settings.

Important: This JSON is a sample to review, not proof that a rule has been applied. Before applying it, check AWS lifecycle eligibility and minimum object-size behavior. The 15-day transition is a separate example from the 30-day rule in Question 2. For versioned buckets, configure noncurrent-version actions if older versions must also be removed.

Conclusion: S3 Lifecycle Management helps automate storage-class transitions and object expiration. Frequently accessed restaurant images can remain in S3 Standard, while older data can move to a lower-cost archival class. Prefix-based rules can also remove expired promotional images automatically.

Practical evidence to attach: (1) Bucket with three uploaded menu images, (2) lifecycle rule for 30-day Glacier transition, (3) AWS Pricing Calculator estimate screenshot, and (4) lifecycle rule for offers/ expiration after 60 days.